

- 1) (a) $p \rightarrow q$ & $q \rightarrow r$ Prove $p \rightarrow r$
 Suppose p , since $p \rightarrow q$, therefore q . Since $q \rightarrow r$ therefore r . Thus $p \rightarrow r$

$p \rightarrow q$
 $q \rightarrow r$
 p
 q
 r

- (b) $\neg R \rightarrow (p \rightarrow \neg q) = \neg R \rightarrow \neg(p \wedge q)$ $p \rightarrow q \rightarrow R$
 $\neg R \rightarrow \neg(p \wedge q)$ R

p

q

Since we know $p \wedge q$. By modus tollens, we conclude R
 (Back we different method by contrapositive).

- 2) (a) Suppose p , $p \rightarrow q$ means q . $r \rightarrow \neg q$ and q means $\neg r$
 Therefore if $p \rightarrow q$ & $r \rightarrow \neg q$, then $p \rightarrow \neg r$.

$p \rightarrow q$ $p \rightarrow \neg r$
 $r \rightarrow \neg q$

p

$\neg r$

q

$\neg r$

- (b) we can use logical equivalences as well. But for me

p $q \rightarrow \neg(q \rightarrow \neg p)$

By contrapositive, assume $q \rightarrow \neg p$. Since we know p & $q \rightarrow \neg p$, we can conclude $\neg q$. Hence proved.

$$3) A \subseteq B \quad x \in A \rightarrow x \notin B$$

B & C are disjoint
means $x \notin B \vee x \notin C$

$$x \in A$$

$$x \notin B$$

This can be proved both by $p \rightarrow q$ & contradiction.

• As $x \in A \rightarrow x \in C$ because $A \subseteq C$. As $x \in C$ & $x \notin B \vee x \notin C$
so $x \notin B$. Proved.

• Suppose $x \in B$. As $x \in A$ & $A \subseteq C$, $x \in C$. As
 $x \notin C \vee x \notin C$ & $x \in B \rightarrow x \notin C$. But this is
contradiction that $x \in C$.

4) This can also be done in both ways. Let us try contradiction.
 $A \setminus B$ is disjoint from C means

$$(x \notin A \vee x \in B) \vee x \notin C$$

Contradiction.

$$x \in A$$

$$x \in C$$

$$x \notin B$$

Suppose $x \notin B$. Since $x \in A$ & $x \notin B$ & $(x \notin A \vee x \in B) \vee x \notin C$,
we conclude $x \notin C$ which is contradiction.

• We should use other methods if they work better than
contradiction.

$\rightarrow$ The suppose $x \in C$. Since $(x \notin A \vee x \in B) \vee x \notin C$ we can
conclude $x \notin A \vee x \in B$. Since $x \in A$, we conclude $x \in B$.
Hence Proved. ($\therefore p \vee q, \neg q : p$)

5) Suppose $x \in A \cap B$ & $x \notin B \cap C$. Means $x \in A$ & $x \notin B$
 & $x \in B$ & $x \notin C$. We have a contradiction
 so it cannot be true that

$$\begin{array}{l} 6) \quad A \cap C \subseteq B \\ \quad a \in C \\ \quad a \in A \end{array} \quad \begin{array}{l} a \notin A \cap B \\ = a \in A \rightarrow a \in B \\ a \in B \end{array}$$

Suppose $a \notin B$. Since $a \in A$ & $a \in C$, $A \cap C$. Since
 $A \cap C \subseteq B$ - so $a \in B$ But this is a contradiction.

$$\begin{array}{l} 7) \quad A \subseteq B \\ \quad a \in A \end{array} \quad a \in C$$

$$a \notin B \cap C = a \in B \rightarrow a \in C$$

Suppose $a \notin C$. As $a \in A$ and $A \subseteq B$, $a \in B$. From
 $a \in B$ & $a \in B \rightarrow a \in C$, $a \in C$ But this is a contradiction.

$$\begin{array}{l} 8) \quad y + x = 2y - x \\ \quad \neg(x=0 \wedge y=0) \\ \quad x \neq 0 \vee y \neq 0 \end{array} \quad y \neq 0$$

$\rightarrow$ Suppose $y=0$. Then for $x \neq 0 \vee y \neq 0$, we can conclude
 that $x \neq 0$. Putting $y=0$ gives $0 + x = 2(0) - x$
 $\rightarrow x = -x$ which itself a contradiction. The only
 value of x that satisfies this is $x=0$. But that is
 contradiction over $x \neq 0$.

9) The key to its solution is that when we suppose something it becomes our given & we can use it to prove.

Suppose $a > 1$, dividing by a , we get $1 > \frac{1}{a}$.

Combining these 2, we get $a > 1 > \frac{1}{a}$ so $a > \frac{1}{a}$ which is contradiction, so $a < 1$. Suppose $a > 0$. Then $a < 1$. Dividing $a < 1$ by a , we get $1 < \frac{1}{a}$ ($\because$ Note that a is between 0 & 1, so a is +ve & direction of inequality is preserved).

Combining $a < 1$ & $1 < \frac{1}{a}$, we get $a < 1 < \frac{1}{a}$. Since $1 < \frac{1}{a}$ & $1 < \frac{1}{a}$, so $1 < \frac{1}{a} < b$, so $b > 1$. Dividing by b , we get $1 > \frac{1}{b}$, combining these, we get $b > 1 > \frac{1}{b}$ which means $b > 1$ which is a contradiction. So $a \neq 0$ but $a \leq 0$.

But since $a \neq 0$, so $a < 0$. Finally suppose $a > -1$, so $-1 < a < 0$. (a is -ve). Dividing by positive number $-a$, we get $\frac{a}{-a} > \frac{-1}{-a} = -1 > \frac{1}{a}$. Since $a > -1$ & $-1 > \frac{1}{a}$, we get $a > -1 > \frac{1}{a}$ so $a > \frac{1}{a}$ which is a contradiction. Hence $a < -1$.

($\because$ we used 3 proofs by contradiction here)

10) Given
 $x, y \in \mathbb{R}$

$$x^2 + y^2 = 2x + y$$

$$y \neq 0$$

$\downarrow$

$$x = 0$$

Goal
 $x \neq 0$

$\downarrow$

$$y = 0$$

using given of problem
to find contradiction

Suppose $x=0$, then $0^2 y = 2(0) + y \rightarrow y=0$ which is a contradiction, so $x \neq 0$ & $y \neq 0$.

i) Given $x, y \in \mathbb{R}$

Goal $y=3$

$x \neq 0$

$$y = (3x^2 + 2y) / (x^2 + 2)$$

Assuming $x \neq 0$, we divide both sides by $x^2 + 2 \neq 0$ get

$$y(x^2 + 2) = 3x^2 + 2y \rightarrow x^2 y + 2y = 3x^2 + 2y$$

$$3x^2 = x^2 y \rightarrow x^2(3 - y) = 0 \quad \text{As } x \neq 0, x^2 \neq 0$$

so $3 - y = 0 \rightarrow y = 3$. which is a direct proof.

If $x \neq 0$ then $x^2 \neq 0$ can improved by contradiction or 8 labels for all x

12) $p \rightarrow q, \neg q \rightarrow \neg p$

p	q	$p \rightarrow q$	$\neg q$	$\neg p$
T	T	T	F	F
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

valid.

15) $p \rightarrow (q \rightarrow r)$ $p \rightarrow q \rightarrow \neg r$ $\rightarrow$ $(p \rightarrow \neg q)$

p	q	r	$p \rightarrow (q \rightarrow r)$	$p \rightarrow q \rightarrow \neg r$	$(p \rightarrow \neg q)$
T	T	T	T	F	F
T	T	F	F	T	F
T	F	T	T	F	T
T	F	F	T	T	T
F	T	T	T	F	F
F	T	F	T	T	F
F	F	T	T	F	T
F	F	F	T	T	T

Exercises

- *1. This problem could be solved by using truth tables, but don't do it that way. Instead, use the methods for writing proofs discussed so far in this chapter. (See Example 3.2.4.)
 - (a) Suppose $P \rightarrow Q$ and $Q \rightarrow R$ are both true. Prove that $P \rightarrow R$ is true.
 - (b) Suppose $\neg R \rightarrow (P \rightarrow \neg Q)$ is true. Prove that $P \rightarrow (Q \rightarrow R)$ is true.
2. This problem could be solved by using truth tables, but don't do it that way. Instead, use the methods for writing proofs discussed so far in this chapter. (See Example 3.2.4.)
 - (a) Suppose $P \rightarrow Q$ and $R \rightarrow \neg Q$ are both true. Prove that $P \rightarrow \neg R$ is true.
 - (b) Suppose that P is true. Prove that $Q \rightarrow \neg(Q \rightarrow \neg P)$ is true.
3. Suppose $A \subseteq C$, and B and C are disjoint. Prove that if $x \in A$ then $x \notin B$.
4. Suppose that $A \setminus B$ is disjoint from C and $x \in A$. Prove that if $x \in C$ then $x \in B$.
- *5. Prove that it cannot be the case that $x \in A \setminus B$ and $x \in B \setminus C$.
- *6. Use the method of proof by contradiction to prove the theorem in Example 3.2.1.
7. Use the method of proof by contradiction to prove the theorem in Example 3.2.5.
8. Suppose that $y + x = 2y - x$, and x and y are not both zero. Prove that $y \neq 0$.
- *9. Suppose that a and b are nonzero real numbers. Prove that if $a < 1/a < b < 1/b$ then $a < -1$.
10. Suppose that x and v are real numbers. Prove that if $x^2y = 2x + y$, then if $y \neq 0$ then $v \neq 0$.
11. Suppose that x and y are real numbers. Prove that if $x \neq 0$, then if $y = (3x^2 + 2v)/(x^2 + 2)$ then $y = 3$. *Answer here: y cannot be given the value 3 because all infinite values of y to show contradiction*
- *12. Consider the following incorrect theorem:

Incorrect Theorem. Suppose x and y are real numbers and $x + y = 10$. Then $x \neq 3$ and $y \neq 8$.

(a) What's wrong with the following proof of the theorem?

Proof. Suppose the conclusion of the theorem is false. Then $x = 3$ and $y = 8$. But then $x + y = 11$, which contradicts the given

as $y = 10 \rightarrow (x \neq 3 \wedge y \neq 8)$. The negation of conclusion is $x = 3 \vee y = 8$ not $x = 3 \wedge y = 8$ which is the problem in proof.

information that $x + y = 10$. Therefore the conclusion must be true. $\square$

- (b) Show that the theorem is incorrect by finding a counterexample.
13. Consider the following incorrect theorem: *$x=3 \wedge y=7$ is counterexample*

Incorrect Theorem. Suppose that $A \subseteq C$, $B \subseteq C$, and $x \in A$. Then $x \in B$.

- (a) What's wrong with the following proof of the theorem?

$x \notin B$ doesn't guarantee that $x \notin C$ by definition of subset. So it's a wrong assumption.

Proof. Suppose that $x \notin B$. Since $x \in A$ and $A \subseteq C$, $x \in C$. Since $x \notin B$ and $B \subseteq C$, $x \notin C$. But now we have proven both $x \in C$ and $x \notin C$, so we have reached a contradiction. Therefore $x \in B$. $\square$

- (b) Show that the theorem is incorrect by finding a counterexample. *$B = \{1, 2, 3\}$
 $C = \{1, 2, 3, 4\}$*
14. Use truth tables to show that modus tollens is a valid rule of inference.
- *15. Use truth tables to check the correctness of the theorem in Example 3.2.4.
16. Use truth tables to check the correctness of the statements in exercise 1.
17. Use truth tables to check the correctness of the statements in exercise 2.
18. Can the proof in Example 3.2.2 be modified to prove that if $x^2 + y = 13$ and $x \neq 3$ then $y \neq 4$? Explain.

3.3. Proofs Involving Quantifiers

Look again at Example 3.2.3. In that example we said that x could be anything at all, and we proved the statement $\forall x (x \in A \rightarrow x \in C)$. We used the word "any" to mean "any element of the universe".

16) (a)

$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	T
F	T	F
F	F	T
F	F	T

Correct.

Rest are same way

18) Given Local

$$x^2 + y = 13$$

$$x \neq 3$$

$$y \neq 4$$

Suppose $y = 4$, then $x^2 + 4 = 13 \rightarrow x^2 = 9, x = \pm 3$
 But $x = -3$ is counterexample to theorem. This means
 according to our statement: $(p \wedge q) \rightarrow r$ & $(p \wedge r) \rightarrow q$
 are different things.